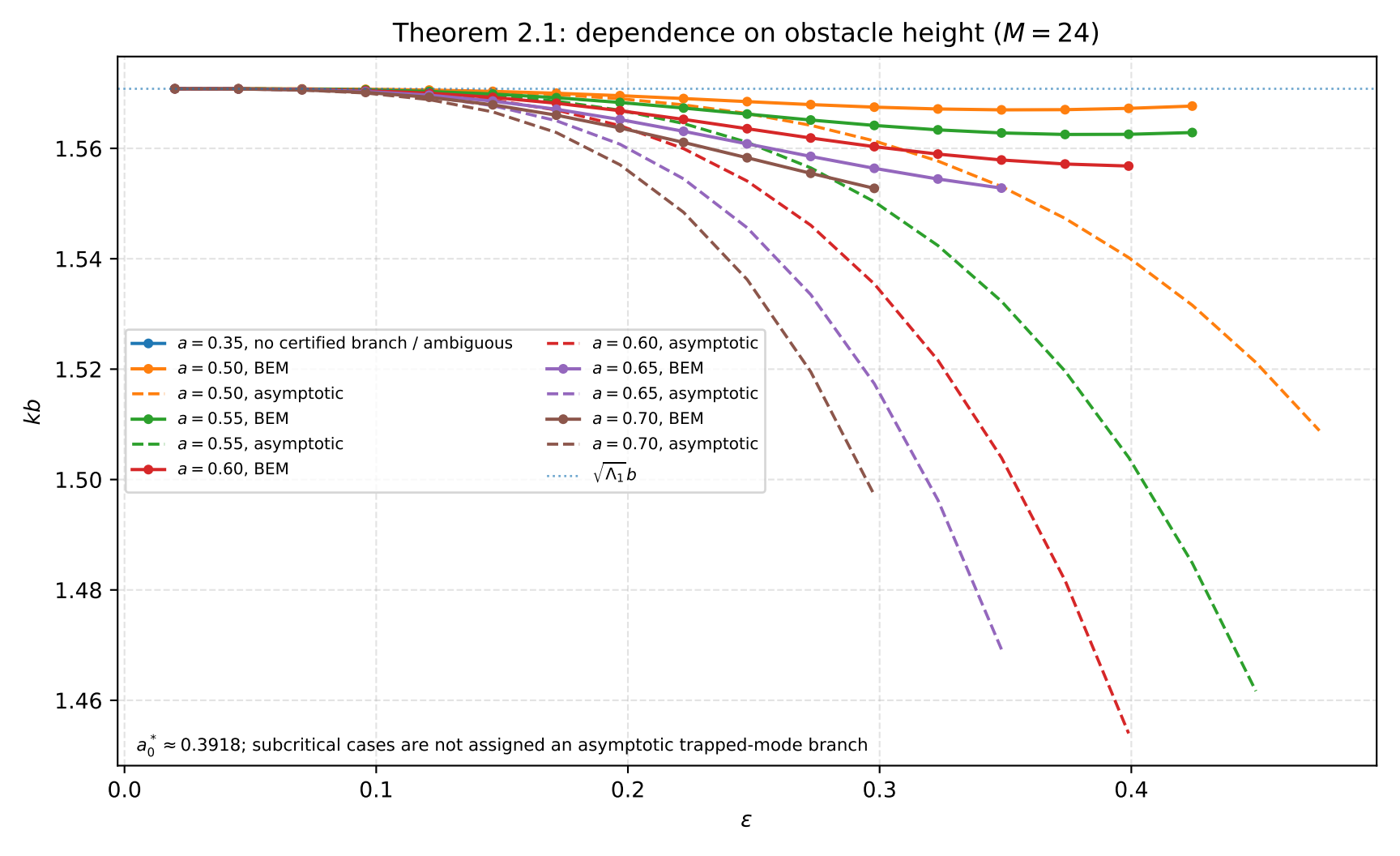
Theorem 2.1: dependence on obstacle height ($M = 24$)
kb
ε
1.46
1.48
1.50
1.52
1.54
1.56
0.0
0.1
0.2
0.3
0.4
$a = 0.35$, no certified branch / ambiguous
$a = 0.50$, BEM
$a = 0.50$, asymptotic
$a = 0.55$, BEM
$a = 0.55$, asymptotic
$a = 0.60$, BEM
$a = 0.60$, asymptotic
$a = 0.65$, BEM
$a = 0.65$, asymptotic
$a = 0.70$, BEM
$a = 0.70$, asymptotic
$\sqrt{\Lambda_1}\, b$
$a_0^* \approx 0.3918$; subcritical cases are not assigned an asymptotic trapped-mode branch